Report · 9 Nov 2026

This year, the highest fire count and Diwali fell on the same day

By Hammad Tariq

Every autumn, two human activities foul the air of both Punjabs: farmers burning crop stubble, and Diwali fireworks. This year they fell on the same day. On Sunday 8 November, Diwali, satellites recorded 5,419 fire detections across Pakistan and northern India, the most of the season so far; in Indian Punjab the count more than tripled. In none of the seven seasons since 2019 had the busiest fire day fallen on Diwali. In Delhi, six of the 13 independent sensors on IQAir passed AQI 1,000 on Diwali night, one reaching 2,089. By 9 am on Monday, Lahore's ground monitors averaged 398.5 µg/m³ of fine particles (PM2.5), a US AQI of 646, against 219 at the same hour on Sunday. All 48 Lahore monitors with readings on both mornings were higher, and two passed AQI 1,000. Multan and Faisalabad went through the same night. The forecast model behind many free air-quality apps had Lahore getting cleaner. Lahore's own evening was quiet; its air worsened after midnight, on a light east wind that had spent Diwali evening over Indian Punjab. The data cannot say how much of Monday morning came from the fires, how much from the fireworks, and how much from Lahore itself.

5,419

satellite fire detections on Diwali, 8 November, the season's highest so far, across Pakistan and northern India

398.5

µg/m³ PM2.5, Lahore's city average at 9 am Monday (AQI 646), against 219 at 9 am Sunday

1,000–2,089

US AQI at six of 13 independent Delhi sensors on Diwali night, against about 200–330 at midday Sunday

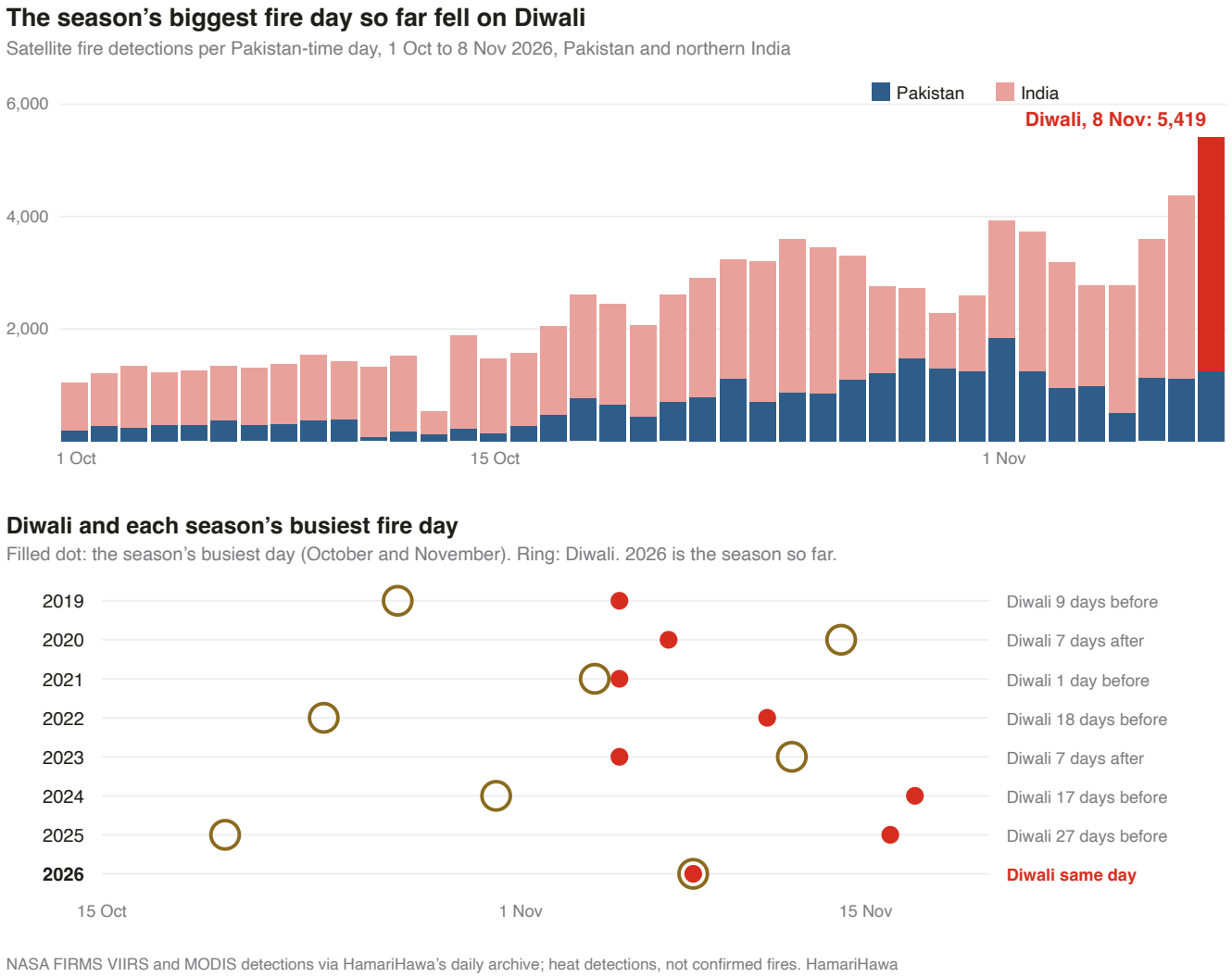
148 → 85

µg/m³, Lahore in the CAMS forecast model, 9 pm to 9 am, while the monitors went from 212 to 398.5

Check this report. Every number here is in `data.json`, with its source, time, place and units. Air readings are IQAir's hourly city averages and station histories, read on the afternoon and evening of 9 November. Fire counts are NASA satellite heat detections, not confirmed fires. The air paths are an approximate guide built from weather-model winds, not a dispersion model. This is a monitoring report: it sets out what the instruments show, and says plainly what they cannot separate.

Two things on one day

Crop burning in the two Punjabs builds through late October and peaks in early to mid November. Diwali moves with the lunar calendar, between mid October and mid November. In the seven seasons from 2019 to 2025 they fell within a day of each other only once: in 2021, Diwali was the day before the busiest fire day. This year they fell on the same day.



Satellite fire detections per Pakistan-time day, NASA FIRMS (VIIRS and MODIS), all land types

Day	Pakistani Punjab	Indian Punjab	Both countries, whole region
Friday 6 Nov	1,050	635	3,595
Saturday 7 Nov	1,029	501	4,371
Sunday 8 Nov (Diwali)	1,143	1,716	5,419

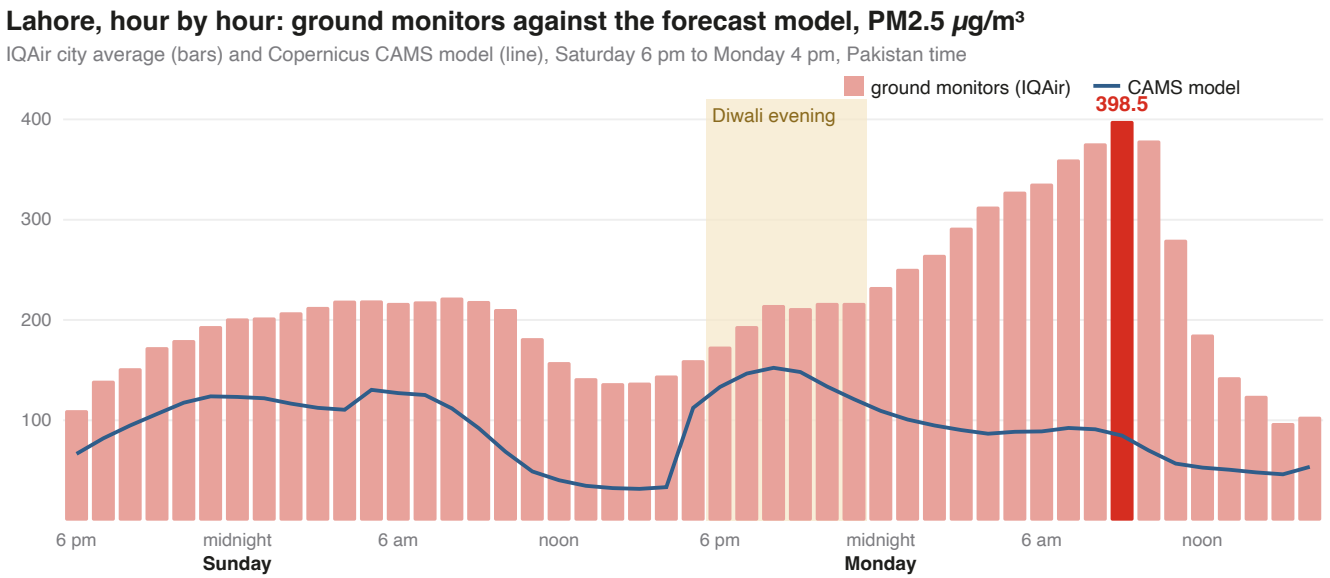
Across Pakistan and northern India, Sunday's 5,419 detections were the most of the season so far. The jump came in Indian Punjab, where detections more than tripled from Saturday, centred on Sangrur (245), Bathinda (213) and Firozpur (207). On the Pakistani side the count held at its usual level of about 1,000 to 1,150 a day, led by Jhang (287) and Hafizabad (193); Pakistan's busiest day so far was 1 November, with 1,849. In 2024 and 2025 the busiest day came in mid November, so Sunday's total may yet be passed. And it is still well below the peaks of 2019–2023, when the busiest day of each season reached 9,500 to 13,800 detections, even though more satellites are counting now.

Satellites count fires, and forecast models take those counts in; nothing in either sees fireworks. What follows is what the ground monitors recorded in Lahore and Delhi, and in two other Pakistani cities, and where the wind took the air in between.

Lahore, hour by hour

Lahore's air was already bad on Sunday. Its city average stayed between 200 and 223 µg/m³ from midnight to 9 am, then fell to 137 in the afternoon as the sun heated the ground and the smoke mixed upward. That is the usual daily rhythm in a Lahore November: worst at night and in the early morning, under a lid of warmer air, and best in mid-afternoon.

Sunday evening was no worse than Sunday morning. Across Lahore's monitors, the median of each one's highest reading between 7 and 11 pm was AQI 314, against a median of 294 at 9 am. The change came after midnight. The city average climbed every hour: 233 µg/m³ at midnight, 313 at 4 am, 398.5 at 9 am, 1.8 times the same hour on Sunday. Then it fell fast: 280 at 11 am, 186 at noon, 97 at 3 pm.

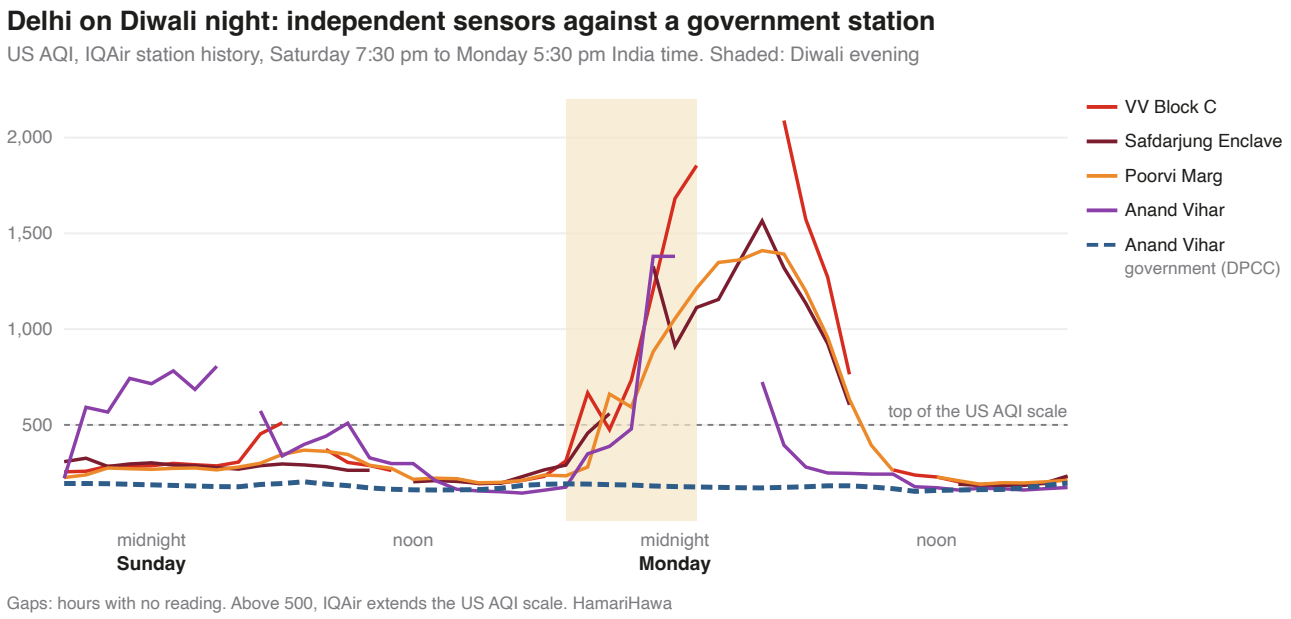


This was not one or two monitors. IQAir's Lahore page links to 51. At 9 am Monday the median one read AQI 651, against 294 on Sunday, and 41 were above 500, the top of the US AQI scale. The highest of the night were Allama Iqbal Town (1,063) and a monitor in the Quaid-e-Azam Industrial Estate (1,012). So the readings above 1,000 that circulated on Monday morning were real, at single monitors; the city's average peaked at 646.

By daily average it was not the season's worst day: IQAir's figure for Monday, up to the afternoon, was AQI 380, against 416 on 2 November, the day after Pakistan's biggest day of fire detections so far this season (1,849 on 1 November). What set 9 November apart was the shape: one very bad night and morning, then a quick fall.

What Diwali did in Delhi

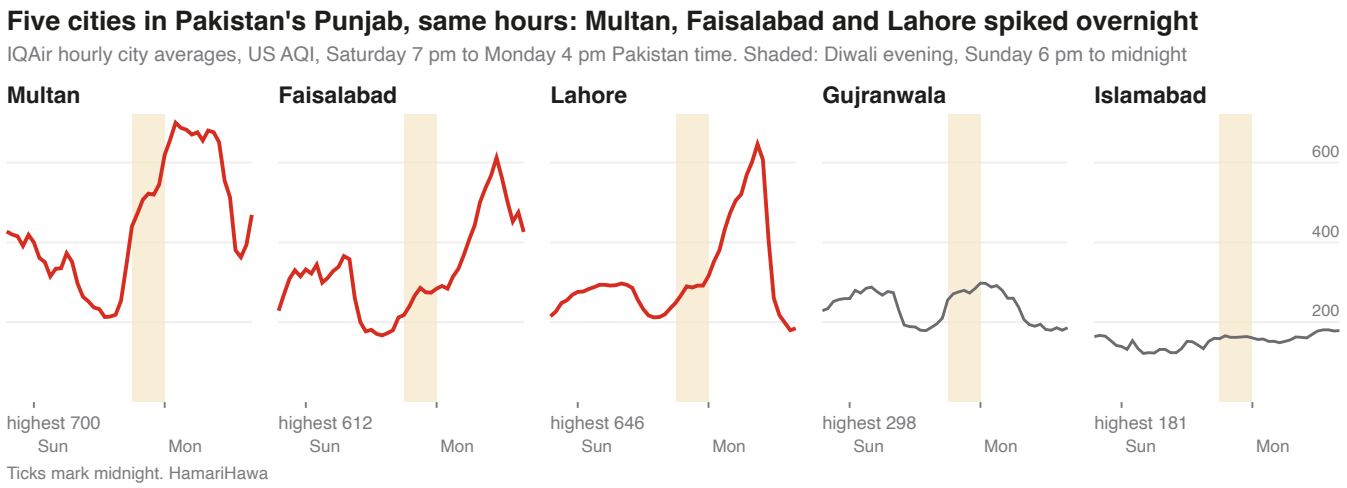
Fireworks do not show up in satellite fire counts, and the emission inventories behind CAMS have no Diwali spike in them. Where monitors are close to the celebrations, though, they are hard to miss. In Delhi, independent sensors that read between about AQI 200 and 330 at midday Sunday climbed through the evening: VV Block C to 1,206 by 10:30 pm and 1,682 by 11:30 pm, Anand Vihar to 1,380, Safdarjung Enclave to 1,328, Vivek Vihar to 1,591. The highest reading of the night, 2,089 at VV Block C, came at 4:30 am. In all, six of the 13 independent sensors IQAir shows in Delhi passed 1,000 during the night; those on the city's outer north-west and west, at Narela, Mundka and Najafgarh, stayed below 400.



The government stations told a different story, at least as IQAir shows them. All 17 Delhi Pollution Control Committee and IITM stations on IQAir read between 174 and 198 in the five hours from 7:30 pm, and many read identically to each other hour after hour: Mundka, Narela, Patparganj, Rohini, Sonia Vihar and Vivek Vihar all show 186, 181, 178, 176, 174. Different stations cannot measure identical air for five hours, so these look like estimates filling in for missing live data. Because they dominate IQAir's Delhi city average, that average showed Delhi getting cleaner through Diwali night. We have not checked this against India's official CPCB portal, and it may be a problem in IQAir's data feed rather than at the stations.

The same caution applies across the border from Lahore. Amritsar and Ludhiana each have a single government station on IQAir, and neither showed a spike, but their series have the same unusually smooth shape. We do not know what Indian Punjab's air did on Diwali night.

Multan and Faisalabad too



Multan's average stayed above AQI 600 from midnight to 10 am, peaking at 700 at 2 am. Faisalabad's was above 500 from 8 am to 1 pm, peaking at 612 at 11 am. Gujranwala, 70 km north of Lahore, peaked at 298, barely above Saturday night's 288, and Islamabad stayed under 190.

What the forecast showed

The Copernicus Atmosphere Monitoring Service (CAMS) model feeds many free air-quality apps, including the forecast on this site. In a Lahore winter it reads low: on Saturday night the city's monitors read 1.5 to 2 times the model's PM2.5. On Sunday night the model and the monitors went in opposite directions. CAMS had Lahore falling from 148 µg/m³ at 9 pm to 85 at 9 am; the monitors rose from 212 to 398.5, 4.7 times the model.

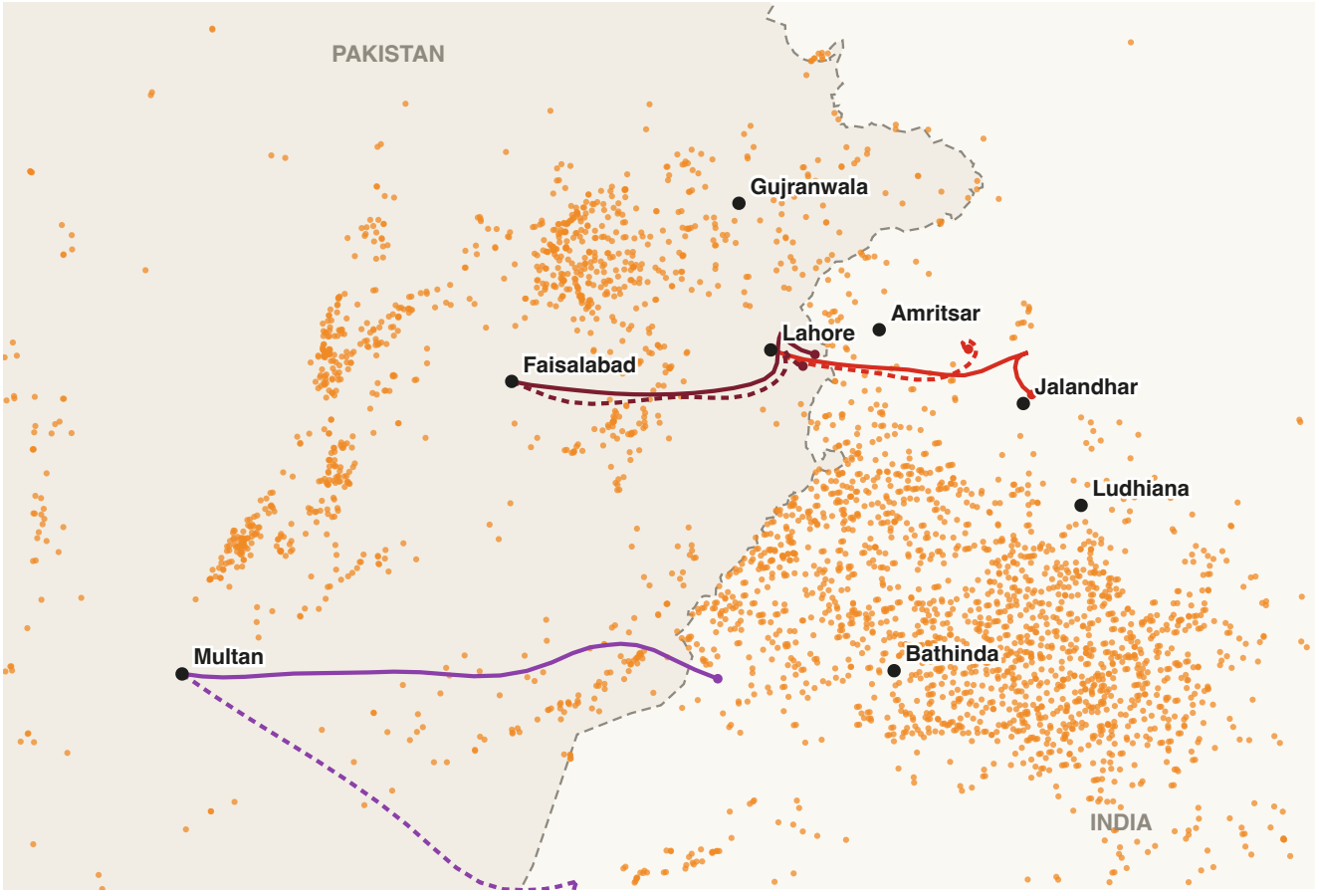
Anyone checking a forecast built on CAMS on Sunday evening would have seen Lahore's air getting better overnight. The model gave no sign of what came.

Where Lahore's air came from

From Saturday evening the wind across both Punjabs turned round from the north-west to the east and stayed light: at Lahore it averaged under 8 km/h through both nights. Following it backwards, hour by hour, shows roughly where each city's Monday-morning air had been.

Where Monday morning's air came from

Approximate 24-hour paths of air arriving at 9 am Monday (near the ground and at about 800 m), over Sunday's fire detections



fire detection, 8 Nov (NASA FIRMS) — path near the ground — path at about 800 m — Lahore · Faisalabad · Multan; dot = 24 h earlier

Paths from GFS winds (Open-Meteo, 0.5°), horizontal only: a guide to direction, not a dispersion model. Dashed line: Pakistan–India boundary. HamariHawa

- **Lahore.** At 6 pm on Sunday, the air that reached Lahore at 9 am was 93 to 115 km to the east, over Kapurthala and eastern Tarn Taran districts. At midnight it was 60 to 73 km away, near the town of Tarn Taran, about 25 km south of Amritsar. It crossed the border at about dawn. Within 20 km of that path, the satellites detected 42 to 47 fires in the 24 hours before the air passed, all but two or three in India, most of them in Tarn Taran, Kapurthala, Amritsar and Jalandhar.
- **Faisalabad.** From noon to 6 pm on Sunday, the air that reached Faisalabad on Monday morning was over Lahore. Overnight it drifted west across Sheikhupura and Nankana Sahib. It passed about 80 fire detections, about nine in ten of them in Pakistan.
- **Multan.** On Sunday evening the air that reached Multan was near Pakpattan, and overnight it crossed Vehari district. Near the ground it passed about 75 fire detections, all but four in Pakistan. A path at about 800 m came instead from Ganganagar in Indian Rajasthan.

The same east wind brought each city something different: Lahore's air came out of Indian Punjab on Diwali night, Faisalabad's out of Lahore and the farmland to its west, Multan's across south Punjab's fields.

What we can and cannot say

We checked the obvious explanations against the weather model and the monitors.

- **Not fire counts alone.** The air that reached Lahore on Sunday morning, from the north and west, had passed over more fires than Monday's air did: about 300 Pakistani detections in Hafizabad and Nankana Sahib along its path at about 800 m. Yet Sunday's 9 am reading was 219, and Monday's 398.5. Monday's air passed 42 to 47, most of them in Tarn Taran, Kapurthala, Amritsar and Jalandhar rather than in Indian Punjab's busiest districts further south.
- **Not stiller air.** In the GFS weather model, Lahore's Sunday night was slightly windier (7.3 km/h on average, against 4.6) and its mixing layer slightly deeper (about 51 m, against 28) than Saturday night's. Both nights had a temperature inversion.
- **Not humidity.** Many of IQAir's Lahore monitors are low-cost sensors that read high in damp air or fog. Lahore's relative humidity stayed below 40% all night.
- **Not Lahore's own evening.** Lahore's monitors did not rise between 7 and 11 pm, when Delhi's did. The rise came after midnight, as the air from the east arrived.

That leaves the things that were different on Sunday night and are missing from every dataset here: the fireworks across Indian Punjab, which the air bound for Lahore passed over between 6 pm and midnight; the smoke of the season's busiest fire day, already in the air over the region that evening; and whatever burned or was emitted along each path that satellites cannot see. Delhi shows what fireworks can do to air close by; we cannot say how much of that reached Lahore 100 km downwind, or how much of Lahore's 398.5 came from fireworks, from crop fires on either side of the border, or from Lahore itself.

What would settle it

Fireworks leave a chemical signature. The metals that give them colour, such as strontium, barium and copper, show up in particle samples and can separate firework smoke from crop smoke and traffic. If the Punjab Environment Protection Department's reference monitors in Lahore, Faisalabad and Multan kept filter samples from 8 and 9 November, analysing them would answer the question this report cannot. A run of a dispersion model such as NOAA's HYSPLIT on the same hours would check the air paths drawn here. And if the province's own forecasts saw this night coming, publishing them hour by hour alongside its monitors would give Lahore the warning the free forecasts did not.

Method and sources

- **Air quality:** IQAir city and station pages, History, Hourly, US AQI and PM2.5, read between 4:30 and 7 pm Pakistan time on 9 November 2026. Each value is for the hour starting at the time shown. Lahore's city average draws on 53 monitors from 35 contributors; 51 Lahore station pages were read individually. Delhi: all 30 station pages listed by IQAir. Lahore's daily averages are IQAir's Daily view tooltips.
- **Fires:** NASA FIRMS near-real-time VIIRS (Suomi NPP, NOAA-20, NOAA-21) and MODIS detections, grouped into Pakistan-time days and districts by HamariHawa's daily archive (/api/day?d=2026-11-08). Season totals and past seasons from /api/days; the season is taken as October and November, and 2026 runs to 8 November.
- **Weather and air paths:** GFS via Open-Meteo, hourly wind at 10 m and at 925 hPa (about 800 m above Lahore) on a 0.5° grid, and mixing-layer height, temperature and humidity at each city. Paths step back every 15 minutes for 24 hours from 9 am Monday; they are horizontal only. Fires were counted within 20 km of a path and in the 24 hours before the air passed that point.
- **Model:** CAMS global PM2.5 via Open-Meteo, grid point nearest central Lahore, as available on 9 November.
- **Diwali:** Sunday 8 November 2026; in past years 27 October 2019, 14 November 2020, 4 November 2021, 24 October 2022, 12 November 2023, 31 October 2024 and 20 October 2025, the main day of Lakshmi Puja.

The extracts behind every chart and table, and the scripts that make them, are in the site's repository under `scripts/reports/lahore-morning-after-diwali/`.

HamariHawa

Live fire and air map of Pakistan and North India, from NASA satellite data.

Made by Hammad Tariq

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